3 DISCUSSION

Visually smooth linear interpolations are often interpreted as a marker of the generative model's performance. Yet this criterion greatly depends on the place where the linear interpolation is performed in latent space, as it can be arbitrarily far off the geodesic (see Figure 2). As a result, we believe considering the whole latent space (e.g. through Monte Carlo sampling) was more statistically meaningful than evaluating several hand-picked interpolations.

From Figure 2 we observe that the "expected worst-case relative improvement" (in this case 10.20%) is non-zero, which shows that the deviation of the straight line from the geodesic is significant enough to take into consideration. Additionally, the standard deviation of the histogram shows how we should be careful in choosing interpolations, as the relative improvement could be vastly different at different locations in latent space.

We present an example strategy of choosing latent linear interpolations in order to compare the interpolation quality of two generative models. We start by randomly sampling pairs of input data points from the test dataset, which we then encode into both latent spaces. In case no encoder is available for the generative model, one can search for closest matching latent points that generate the input samples, as shown in (Lei et al., 2019) for instance. By finding a shorter curve connecting the pairs of points, we can compute relative length improvements for each of the two generators. Then, we choose the pair of points that has the most similar relative improvement, which can be seen as them being located in a similar metric neighborhood (see results in Appendix D). This choice rules out the comparisons between the best linear interpolation of one model to the worst of the other.

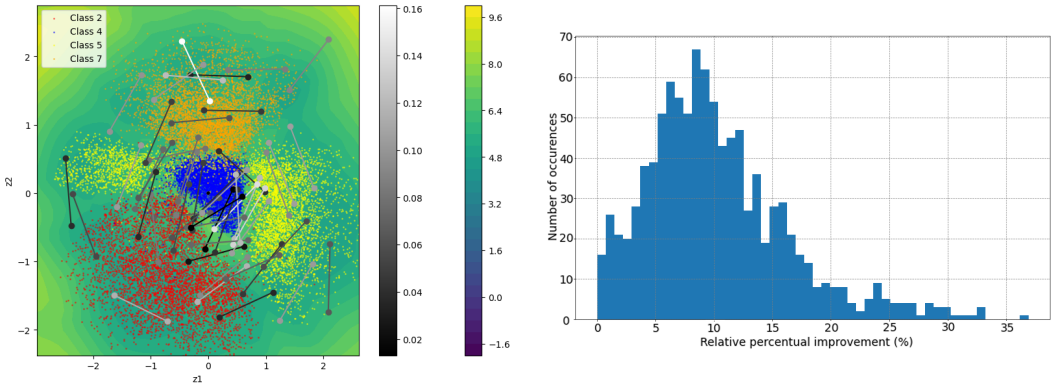

Figure 2: The latent space of this VAE is 2D. *Left*: In the background $\log \sqrt{\det \mathbf{M}_z}$ is plotted in color, together with clusters of encoded MNIST test data (digits 2,4,5,7). In the foreground we see straight lines connecting sampled pairs of points as described by Algorithm 1, where the brightness indicates how much the curve length can be shortened (0.14 indicates that the shorter curve is 14% shorter than the straight line Riemannian length). *Right*: Histogram of the distribution of relative length improvements using 1000 samples. A larger version can be found in Appendix C.

4 CONCLUSION

Accounting for the non-Euclidean nature of the latent space of generative models, we present a geometry-aware method for indicating the comparability of latent linear interpolations. Through random sampling, an average measure can be obtained over the whole latent space. A limitation of our current implementation is that we do not find global length minimizers/geodesics. Our computation for shorter curves could also be improved (as in Arvanitidis et al. (2019)), mainly due to the expensive computation of the Jacobian. We showed that the latent space geometry cannot be disregarded while evaluating generative models, and we believe geometry may further help us understand deep learning in the future.

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A CUBIC B-SPLINE

The implementation of the cubic B-spline is as follows:

$$
\begin{aligned}
N_{i,1}(t) &= \begin{cases} 1 & \text{for } t_i \leq t < t_{i+1} \\ 0 & \text{otherwise} \end{cases} \\
N_{i,k}(t) &= \frac{t - t_i}{t_{i+k-1} - t_i} N_{i,k-1}(t) + \frac{t_{i+k} - t}{t_{i+k} - t_{i+1}} N_{i+1,k-1}(t) \\
\boldsymbol{C}(t) &= \sum_{i=0}^{n} N_{i,4}(t) \boldsymbol{P}_i \\
\boldsymbol{T} &= (\underbrace{0,0,0,0}_{4}, \underbrace{knots(]0,1[)}_{n+1-4}, \underbrace{1,1,1,1}_{4})
\end{aligned}
$$

Here $N_{i,k}$ are the basis functions, $\boldsymbol{P}_i$ are the control points and $\boldsymbol{T}$ is the knot vector. We want end-point interpolation, hence we have the knot multiplicities above for start and end point. This b-spline has $n+1$ control points and order 4. Accordingly the first derivative is defined as:

$$
\frac{\partial \boldsymbol{C}(t)}{\partial t} = 3 \sum_{i=0}^{n-1} N_{i+1,3}(t) \frac{\boldsymbol{P}_{i+1} - \boldsymbol{P}_i}{T_{i+4} - T_{i+1}}
$$

The knot vector consists of a middle part with $n-3$ elements, which is adjusted everytime a new control point is added. We wish to add new control points in such a way that the old curve geometry is preserved. The implementation (choosing where to put new knots first) is as follows: we find the largest knot interval, and place a new knot in the exact middle of it. Knowing the new knot, we insert a new control point accordingly by adjusting 2 old ones ($i \in [j-2, j]$ where j is the knot index of the new knot) such that curve geometry is kept the same:

$$
\begin{aligned}
\boldsymbol{P}_i^{new} \longleftarrow & (1 - a_i) \boldsymbol{P}_{i-1}^{old} + a_i \boldsymbol{P}_i^{old} \\
a_i = & \frac{t_{new} - t_i}{t_{i+k-1} - t_i}
\end{aligned}
$$

We update from largest i to smaller ones, so we always use the old values and do all operations in-place. t_i are the knots. The point $\boldsymbol{P}_j^{new}$ is not updated in the old control point list, but will instead be our new control point that we insert at position j in the list.

B EFFECT OF LOGISTIC REGRESSION FEATURE MAPPING

As the logistic regression maps an image to probabilities of every class, finding a geodesic in such a pull-back metric is equivalent to travelling fastest between classes, i.e. traversing as few classes as possible. The effect can be observed in Figure 3, where the middle sequence is without any feature mapping, and the bottom one has the logistic regression, skipping over several digits in between.

Figure 3: Interpolating between two fixed points in latent space of a VAE trained on MNIST digits. Top sequence is along the straight line, middle is shorter distance as defined by the pull-back metric of the generator, and bottom is according to metric with a logistic regression feature mapping on top of generator.